확률·랜덤프로세스 기말대비 6/10 (D4) · 1차 — Lecture 08 상관 · 반복기댓값 · 전분산

① 오답노트.md 어제 오답부터 재풀이 ② 아래 신규 문제는 '상한'(시간 부족 시 신규만 이월) ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- 상관계수 $\rho = Cov/(\sigma_X \sigma_Y) \in [-1,1]$ · Cauchy-Schwarz $|E[XY]| \le \sqrt{E[X^2]E[Y^2]}$
- 반복기댓값 $E[X]=E[E[X|Y]]$ · 전분산 $Var(X)=E[Var(X|Y)]+Var(E[X|Y])$ · 랜덤합

문항 목차 (총 11문항)

[1] L08 · IPSRP 5.4.36 — Further2 상관·전분산
[2] L08 · IPSRP 5.4.37 — Further2 상관·전분산
[3] L08 · PSP 7.5.1 — Further2 상관·전분산
[4] L08 · PSP 7.5.2 — Further2 상관·전분산
[5] L08 · PSP 7.5.3 — Further2 상관·전분산
[6] L08 · PSP 7.5.4 — Further2 상관·전분산
[7] L08 · PSP 7.5.5 — Further2 상관·전분산
[8] L08 · PSP 7.6.2 — Further2 상관·전분산
[9] L08 · PSP 7.6.3 — Further2 상관·전분산
[10] 복습 L03 · PSP 3.6.1 — 이산RV1
[11] 복습 L04 · PSP 5.2.1 — 이산RV2

Problem 36

Let X and Y be jointly normal random variables with parameters $\mu_X = -1$, $\sigma_X^2 = 4$, $\mu_Y = 1$, $\sigma_Y^2 = 1$, and $\rho = -\frac{1}{2}$.

a. Find $P(X + 2Y \leq 3)$.
b. Find $Cov(X - Y, X + 2Y)$.

— 풀이 —

Problem 37

Let X and Y be jointly normal random variables with parameters $\mu_X = 1$, $\sigma_X^2 = 4$, $\mu_Y = 1$, $\sigma_Y^2 = 1$, and $\rho = 0$.

a. Find $P(X + 2Y > 4)$.
b. Find $E[X^2 Y^2]$.

— 풀이 —

7.5.1 ● X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2 & 0 \le y \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the PDF $f_Y(y)$, the conditional PDF $f_{X|Y}(x|y)$, and the conditional expected value $\mathrm{E}[X|Y=y]$.

— 풀이 —

7.5.2● Let random variables X and Y have joint PDF $f_{X,Y}(x,y)$ given in Problem 7.5.1. Find the PDF $f_X(x)$, the conditional PDF $f_{Y|X}(y|x)$, and the conditional expected value $E[Y|X=x]$.

— 풀이 —

7.5.3 The probability model for random variable A is

$$P_A(a) = \begin{cases} 1/3 & a = -1, \\ 2/3 & a = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The conditional probability model for random variable B given A is:

$$P_{B|A}(b|-1) = \begin{cases} 1/3 & b = 0, \\ 2/3 & b = 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$P_{B|A}(b|1) = \begin{cases} 1/2 & b = 0, \\ 1/2 & b = 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is the probability model for random variables A and B? Write the joint PMF $P_{A,B}(a,b)$ as a table.

(b) If $A = 1$, what is the conditional expected value $\mathrm{E}[B|A = 1]$?

(c) If $B = 1$, what is the conditional PMF $P_{A|B}(a|1)$?

(d) If $B = 1$, what is the conditional variance $\mathrm{Var}[A|B = 1]$ of A?

(e) What is the covariance $\mathrm{Cov}[A, B]$?

— 풀이 —

7.5.4 For random variables A and B given in Problem 7.5.3, let $U = \mathrm{E}[B|A]$. Find the PMF $P_U(u)$. What is $\mathrm{E}[U] = \mathrm{E}[\mathrm{E}[B|A]]$?

— 풀이 —

7.5.5 Random variables N and K have the joint PMF

$$P_{N,K}(n,k) = \begin{cases} \frac{100^n e^{-100}}{(n+1)!} & \substack{n=0,1,\ldots;\\ k=0,1,\ldots,n,} \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the marginal PMF $P_N(n)$ and the conditional PMF $P_{K|N}(k|n)$.

(b) Find the conditional expected value $\mathrm{E}[K|N=n]$.

(c) Express the random variable $\mathrm{E}[K|N]$ as a function of N and use the iterated expectation to find $\mathrm{E}[K]$.

— 풀이 —

7.6.2 X and Y are jointly Gaussian random variables with $E[X] = E[Y] = 0$ and $\mathrm{Var}[X] = \mathrm{Var}[Y] = 1$. Furthermore, $E[Y|X] = X/2$. Find $f_{X,Y}(x, y)$.

— 풀이 —

7.6.3 A study of bicycle riders found that a male cyclist's speed X (in miles per hour over a 100-mile "century" ride) and weight Y (kg) could be modeled by a bivariate Gaussian PDF $f_{X,Y}(x,y)$ with parameters $\mu_X = 20$, $\sigma_X = 2$, $\mu_Y = 75$, $\sigma_Y = 5$ and $\rho_{X,Y} = -0.6$. In addition, a female cyclist's speed X' and weight Y' could be modeled by a bivariate Gaussian PDF $f_{X',Y'}(x',y')$ with parameters $\mu_{X'} = 15$, $\sigma_{X'} = 2$, $\mu_{Y'} = 50$, $\sigma_{Y'} = 5$ and $\rho_{X',Y'} = -0.6$. For men and women, the negative correlation of speed and weight reflects the common wisdom that fast cyclists are thin. As it happens, cycling is much more popular among men than women; in a mixed group of cyclists, a cyclist is a male with probability $p = 0.80$.

You suspect it's OK to ignore the differences between men and women since for both groups, weight and speed are negatively correlated, with $\rho = -0.6$. To convince yourself this is OK, you decide to study the speed $\hat{X}$ and weight $\hat{Y}$ of a cyclist randomly chosen from a large mixed group of male and female cyclists. How are $\hat{X}$ and $\hat{Y}$ correlated? Explain your answer.

— 풀이 —

3.6.1 Given the random variable Y in Problem 3.4.1, let $U = g(Y) = Y^2$.

(a) Find $P_U(u)$.

(b) Find $F_U(u)$.

(c) Find $\mathrm{E}[U]$.

— 풀이 —

5.2.1 ● Random variables X and Y have the joint PMF

$$P_{X,Y}(x,y) = \begin{cases} cxy & x = 1,2,4; \quad y = 1,3, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is the value of the constant c?

(b) What is $\mathrm{P}[Y < X]$?

(c) What is $\mathrm{P}[Y > X]$?

(d) What is $\mathrm{P}[Y = X]$?

(e) What is $\mathrm{P}[Y = 3]$?

— 풀이 —